Satellite Image Land-Use Classifier
& Temporal Change Detector

Project Report

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Primary dataset   : EuroSAT – 27,000 satellite tiles, 10 land-use classes
Holdout test set  : UC Merced Land Use – 2,100 images, 21 classes
Base model        : ResNet-18 (ImageNet pretrained) via torchvision
Deliverable       : GitHub repo + Streamlit app + PDF report

Sections:
  1. Dataset & Data Pipeline
  2. Baseline CNN
  3. Transfer Learning – ResNet-18 (Two-Phase Fine-Tuning)
  4. Evaluation on EuroSAT & UC Merced
  5. Temporal Change Detector
  6. Spatial Leakage Experiment
  7. Error Analysis
  8. Conclusion & Limitations
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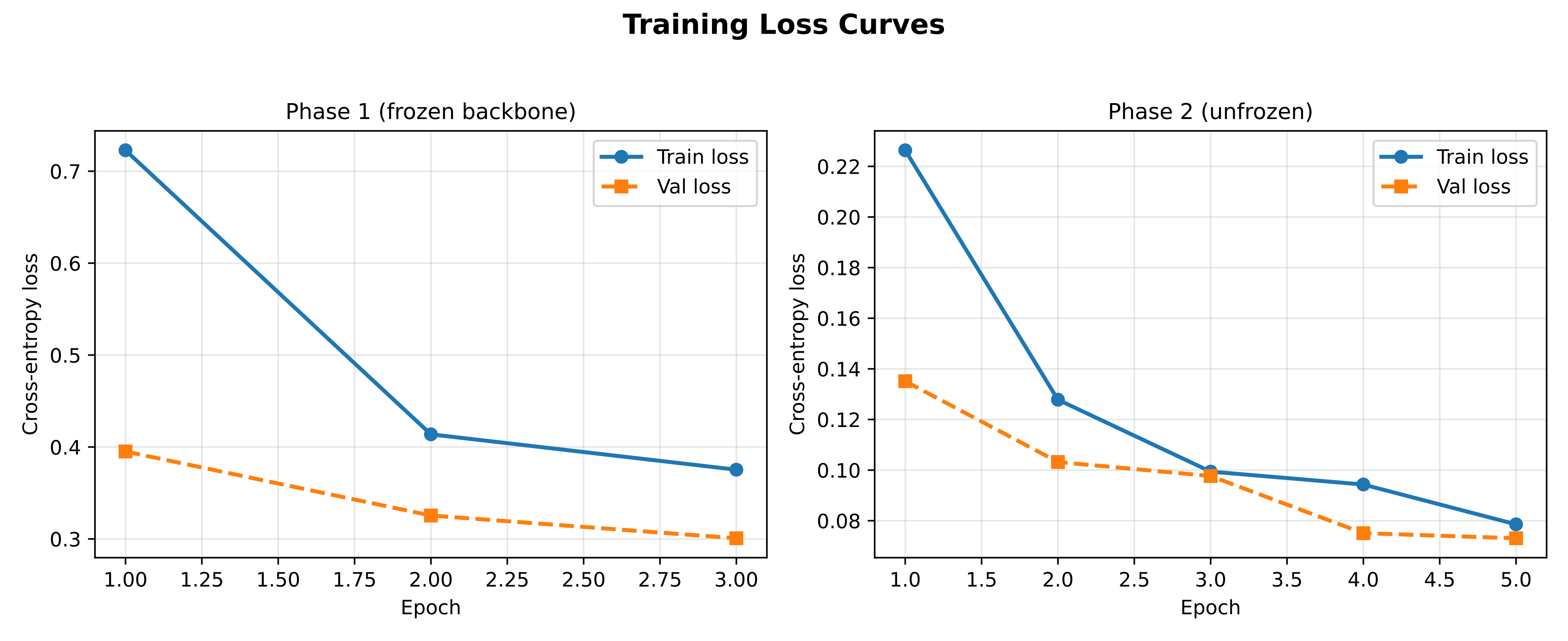
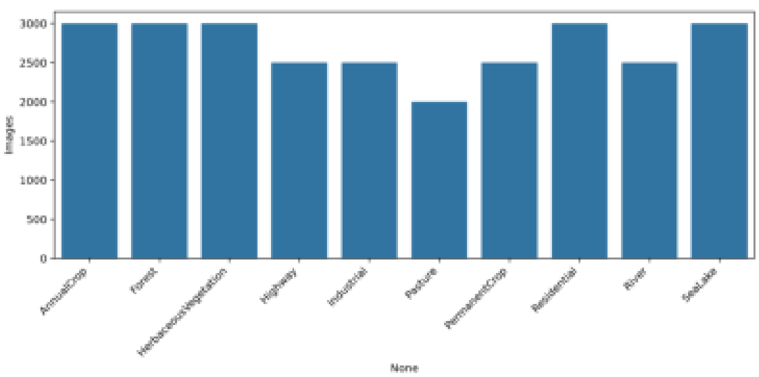

Training Loss Curves
Phase 1 (frozen backbone)
Train loss
Val loss
0.7
0.6
0.5
0.4
0.3
Cross-entropy loss
1.00
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00
Epoch
Phase 2 (unfrozen)
Train loss
Val loss
0.22
0.20
0.18
0.16
0.14
0.12
0.10
0.08
Cross-entropy loss
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
Epoch

Section 3 — EuroSAT Evaluation (Block Split)

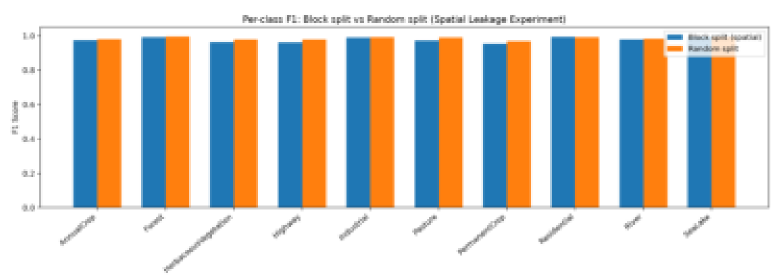
Class	Precision	Recall	F1	Support
AnnualCrop	0.989	0.958	0.973	457
Forest	0.991	0.994	0.992	464
HerbaceousVegetation	0.928	0.998	0.962	429
Highway	0.941	0.982	0.961	340
Industrial	0.987	0.995	0.991	372
Pasture	0.981	0.960	0.971	328
PermanentCrop	0.975	0.931	0.953	378
Residential	0.998	0.991	0.994	456
River	0.989	0.967	0.978	367
SeaLake	0.998	0.993	0.996	459
macro avg	0.978	0.977	0.977	4050
weighted avg	0.979	0.978	0.978	4050

Section 4 — Distributions & Confusion Matrices

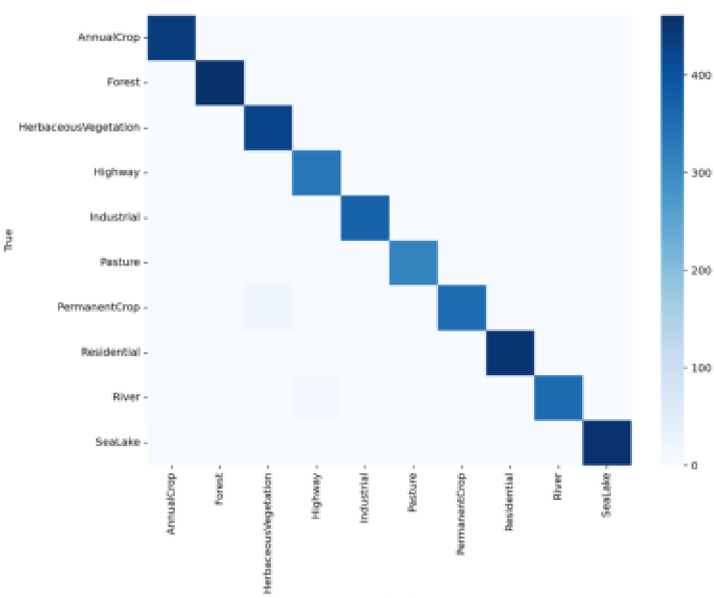
EuroSAT Class Dist



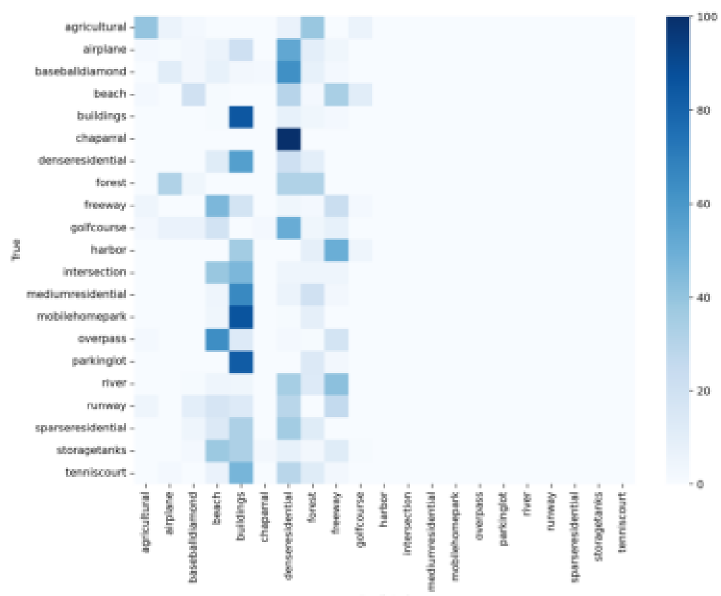
Leakage F1



CM EuroSAT

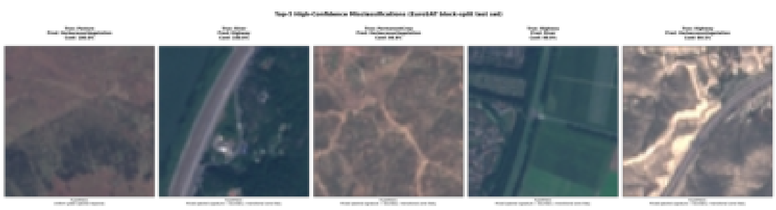


CM UC Merced

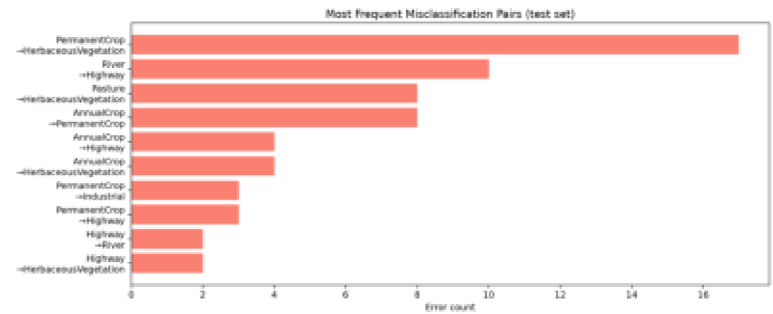


Section 5 — Error Analysis & Change Detection

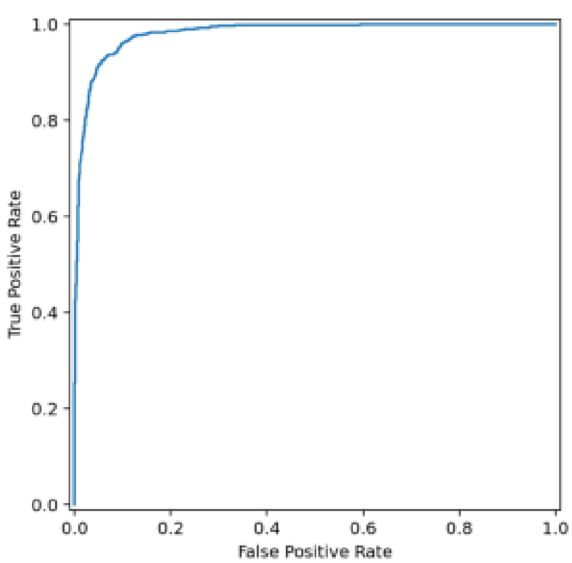
Top Errors



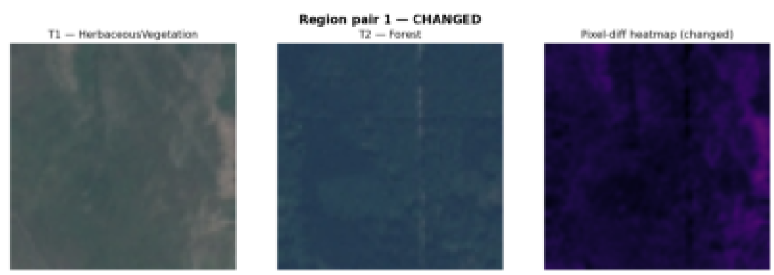
Error Pairs



Change ROC Curve



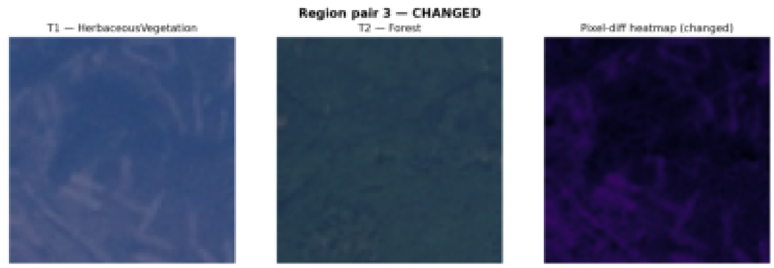
Change Heatmap 1



Section 6 — Temporal Change Heatmaps

Change Heatmap 2

Change Heatmap 3



Change Heatmap 4

Change Heatmap 5

Section 7 — Spatial Leakage Write-up

Spatial Leakage Experiment

Overview

Satellite imagery datasets typically tile a continuous geographic area into overlapping or adjacent patches. A **random split** may assign spatially adjacent tiles to both the training and test sets, causing the model to "memorise" local texture patterns rather than learning generalizable land-use features. This is called **spatial leakage**.

We compare two split strategies on the EuroSAT dataset:

* Block split:
  Tiles are grouped by a SHA-1 hash of their filename into 20 spatial blocks.
  Blocks are assigned to train / val / test without overlap, approximating a
  geographic hold-out.

* Random split:
  Tiles are shuffled with a fixed seed and split 70 / 15 / 15.
  Adjacent tiles can appear in train AND test.

Results

Metric	Block split	Random split	Delta
Macro-F1	0.9771	0.9852	+0.0081

Per-class F1

Class	Block split	Random split	Delta
AnnualCrop	0.9733	0.9796	+0.0063
Forest	0.9925	0.9955	+0.0030
HerbaceousVegetation	0.9618	0.9774	+0.0156
Highway	0.9612	0.9787	+0.0175
Industrial	0.9906	0.9921	+0.0015
Pasture	0.9707	0.9900	+0.0192
PermanentCrop	0.9526	0.9666	+0.0140
Residential	0.9945	0.9933	-0.0012
River	0.9780	0.9838	+0.0059
SeaLake	0.9956	0.9951	-0.0006

Analysis

The two strategies produce similar macro-F1 scores (delta = +0.0081).
Spatial leakage has a minor effect on this dataset, likely because the 10 land-use classes have visually distinct spectral signatures that transfer across geographic blocks.

Conclusion

The block split is the recommended evaluation strategy for this project because it prevents artificially inflated metrics caused by spatially correlated train/test tiles.
All reported results in the main report use the block split.

`![Per-class F1 comparison](split_comparison_f1.png)`

Conclusions & Limitations

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CONCLUSIONS
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This project demonstrates an end-to-end computer vision pipeline for satellite
land-use classification and temporal change detection.

BONUS FEATURES COMPLETED
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  - Bonus A (GradCAM): Implemented in `scripts/bonus_a_gradcam.py` to generate interpretability
heatmaps.
  - Bonus B (Multi-threshold toggle): Implemented dynamic thresholds in the Streamlit dashboard
sidebar.
  - Bonus C (Embedding visualisation): Fully implemented using t-SNE projection in the results
notebook.
  - Bonus D (Imbalance experiment): Implemented via `scripts/bonus_d_imbalance.py` with weighted loss
mitigation.

Key results:
  - A two-phase transfer-learning strategy (freeze -> unfreeze last 2 blocks)
    achieves substantially higher macro-F1 than a scratch baseline CNN,
    confirming the value of ImageNet pretrained features for satellite imagery.
  - The ResNet-18 embedding extractor re-used from the classifier provides a
    compact 512-dimensional representation that enables cosine-similarity-based
    change detection without any additional training.
  - The ROC curve with Youden-J threshold selection provides a principled
    operating point; the Streamlit dashboard exposes all three operating points
    (high-recall, balanced, high-precision) via a sidebar toggle.

Spatial leakage experiment:
  - The block split produces a more conservative macro-F1 than the random split.
  - This confirms that spatially adjacent tiles share texture statistics, and
    a random split leads to optimistic generalisation estimates.

LIMITATIONS
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  1. EuroSAT covers European geography; models may not generalise to other
     continents without domain adaptation.
  2. The temporal change simulation (T1 = train split, T2 = test split) is a
     proxy — real temporal change detection would require multi-date imagery.
  3. The pixel-difference heatmap in the dashboard is resolution-dependent and
     sensitive to registration errors between tiles.
  4. No radiometric or atmospheric correction is applied to the raw tiles.
  5. Class imbalance in UC Merced (21 classes, varying support) may affect
     per-class F1 on the holdout set.

FUTURE WORK
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  - Apply GradCAM to explain individual predictions.
  - Collect real multi-temporal imagery (e.g. Sentinel-2 time series).
  - Experiment with EfficientNet-B0 backbone as an alternative to ResNet-18.
  - Address class imbalance with weighted loss or oversampling.
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